SSEE in the Strong-Gravity Regime: Disformal Geodesics, MIRA Emergence, and Solar-System Compliance

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Abstract

Status note (2026-08). An earlier version of this abstract justified the canonical prediction below by appealing to Paper 6, which fixed a ϕ-dark-matter particle mass and argued that this settled the physical interpretation. **Paper 6 has since retracted that particle**: the subtraction defining its density mixed a measured density with an equation-of-state number, and the S_8 tension it was built to close does not exist in the raw data.

That justification is therefore withdrawn — but the prediction is not, and it is worth being precise about why. Selecting the canonical Bellini–Sawicki limit (b) over the pedagogical limit (a) requires two things, and *neither of them was ever supplied by the particle*:

1. **That the model carries real, clustering cold dark matter.** It does: $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s = 0.119514$, an algebraic forward identity closed under OP-8 in the ω_m-direct reframe. This is ordinary cold dark matter and has no connection to the retracted ϕ-DM sector, which was always an *additional* layer on top of it.
2. **That the residual fifth force is suppressed.** This comes from Paper 7, not Paper 6: $\alpha_B = \alpha_M = \alpha_T = 0$ are derived, and the Bellini–Sawicki effective Newton constant is $\mu - 1 = (\alpha_B + \alpha_M)^2/\alpha_K = 0$ identically.

Both legs stand. The canonical strong-lensing prediction is therefore retained as *unconditional*, now resting on ω_c (OP-8) and the Paper 7 EFT structure rather than on a particle. The two-limit analysis is kept in full below because limit (a) remains pedagogically useful as a falsifiable contrast, not because the choice between them is open.

A scale-dependent two-sector discussion also appeared here (contributions on either side of a free-streaming scale k_{fs}); that is withdrawn with the particle, since the matter sector is single and unpartitioned. The identification of parameter regions where deviations from ΛCDM may become observable with HST/JWST strong-lensing surveys is unaffected.

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1 Introduction

The accelerated expansion of the Universe (Riess et al., 1998; Perlmutter et al., 1999) motivates modifications to the concordance ΛCDM model whose parameters are precisely constrained by Planck CMB data (Planck Collaboration, 2020). Papers 1–7 of the SSEE series established the cosmological background (Almeida Vallejo, 2025a), Bayesian validation (Almeida Vallejo, 2025b), CMB confrontation (Almeida Vallejo, 2025c), algebraic CMB derivation (Almeida Vallejo, 2025d), Israel–Stewart perturbation theory (Almeida Vallejo, 2026a), the growth sector on raw data (Almeida Vallejo, 2026b), and the canonical EFT formulation (Almeida Vallejo, 2026c). All observable predictions in those papers operate within the linear or quasi-linear regime.

The disformal coupling that defines SSEE has observable consequences in the strong-gravity regime that have not yet been examined. In coupled dark-energy models with disformal metrics, two qualitatively new effects arise beyond the linearised regime:

1. **Enhanced deflection angle** — photons propagating through the modified gravitational potential of disformally-coupled dark matter experience a different total deflection relative to pure GR.

2. **Local GR compliance** — any fifth force must vanish on solar-system and stellar scales; in SSEE this follows from selective coupling and the EFT structure, not from a screening radius.

In this paper we derive both effects analytically and obtain numerical predictions using the SSEE algebraic constants. We work explicitly in two limits: (a) an *alternative MOND-like limit* (§4– 5) in which the dark-matter distribution is taken to be sourced entirely by the baryonic seed via the scalar force — here the lensing factor is $\sqrt{\mathcal{A}} \approx 1.9995$, derived directly from the disformal null geodesic; and (b) the *canonical limit* (§6, §6.3) corresponding to the actual SSEE model: a single unpartitioned matter component $\Omega_m = \omega_m/h^2 = 0.308881$, of which the cold dark matter is $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ (an algebraic forward identity, OP-8), together with the EFT Bellini–Sawicki structure $\alpha_B = \alpha_M = \alpha_T = 0$, in which the fifth-force is suppressed to $F_\phi/F_N \sim 10^{-15}$ at sub-Mpc scales and the canonical observable lensing reduces to GR-with-DM. The two limits make incompatible physical assumptions; only limit (b) describes the canonical SSEE model of Papers 1, 6, 7, and 10. We retain limit (a) as a derivation of pedagogical and historical interest, not as a canonical prediction. The $\sqrt{\mathcal{A}} \approx \mathcal{M}$ numerical *near-coincidence* (gap 0.03% because both quantities round to 2) is therefore not promoted to a structural identity: $\sqrt{\mathcal{A}}$ and $\mathcal{M} = \mathcal{A}/2$ are distinct algebraic operations on $\mathcal{A}$ that happen to land near 2. The identities that *do* use $\mathcal{M} = \mathcal{A}/2$ exactly are purely algebraic:

$$\mathcal{M} \;=\; \frac{3\varphi + \pi}{4} \;=\; \frac{\beta_c}{2} \;=\; \frac{\mathcal{A}}{2}\,, \tag{1}$$

where $\beta_c \equiv \mathcal{A} = (3\varphi + \pi)/2$ is the disformal coupling constant fixed in Paper 7. Following the 2026 ω_m-direct reframe (OP-8 dissolved), $\mathcal{M}$ is realised physically as the Hubble *screening amplitude* of Paper 9 ($f^{\mathrm{IR}}_{\mathrm{screen}} = s^{\mathrm{IR}}_K/(3\,\mathcal{M})$). A ratio $\Omega_m/(1+w_0) = 0.308881/0.160050 \approx 1.93$ was also noted here as *numerically close* to $\mathcal{M}$, and was explicitly *not* identified with it. It is now withdrawn as a comparison altogether: the denominator $1 + w_0$ is an equation-of-state parameter, not a second matter density, so the ratio compares two quantities of different kind and its near-2 value carries no content (Paper 1, §1.4, where the "Two-Ω_m" framing is dissolved). In the lensing sector the

quantity naturally appearing in the alternative MOND-like derivation is $\sqrt{\mathcal{A}}$, not $\mathcal{M}$, and the canonical limit (b) recovers GR lensing — see Section 5 for the explicit naming clarification and Section 6.3 for the canonical suppression.

The paper is organised as follows. Section 2 recalls the relevant EFT ingredients from Paper 7. Section 3 derives the quasi-static scalar EOM and gradient closure in the non-relativistic limit. Section 4 constructs the disformal null geodesic and derives the photon deflection angle. Section 5 shows algebraically how $\mathcal{M}$ emerges as the lensing amplification factor. Section 6 shows how solar-system compliance follows from selective coupling and the EFT structure. Section 7 translates the results into strong-lensing observables and identifies JWST/HST falsification windows. Section 8 collects the pre-committed falsifiable predictions. Section 9 discusses the scale dependence and caveats. Section 10 concludes.

2 EFT Foundation: Action and Disformal Coupling

We adopt the SSEE canonical action from Paper 7 (Almeida Vallejo, 2026c):

$$S \;=\; \int d^4x\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R \;+\; P(X,\phi)\right] \;+\; S_{\rm DM}[\tilde{g}_{\mu\nu};\,\psi_{\rm DM}] \;+\; S_b[g_{\mu\nu};\,\psi_b]\,, \tag{2}$$

where $X = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$ is the canonical kinetic invariant, and the k-essence kernel is

$$P(X,\phi) \;=\; \frac{X}{\mathrm{KAL}_0} + \frac{X^2}{M^4} - V_0\, e^{-\lambda\phi/M_{\rm Pl}}\,. \tag{3}$$

Action (2) is a k-essence model within the minimal Horndeski class (Horndeski, 1974; Crisostomi & Koyama, 2018) ($G_3 = G_4 - M_{\rm Pl}^2/2 = G_5 = 0$). Baryons ($\psi_b$) couple to the physical metric $g_{\mu\nu}$; dark matter ($\psi_{\rm DM}$) couples to the disformal metric (Bekenstein, 1993; Zumalacarregui & Garcia-Bellido, 2014)

$$\tilde{g}_{\mu\nu} \;=\; g_{\mu\nu} + \frac{2}{M^4}\,\partial_\mu\phi\,\partial_\nu\phi\,. \tag{4}$$

The algebraic constants fixed by φ and π (Paper 7) are:

$$\mathrm{KAL}_0 = \beta + \pi = \frac{\varphi+\pi}{2} + \pi \approx 5.521\,, \tag{5}$$

$$\beta_c \equiv \mathcal{A} = \frac{3\varphi+\pi}{2} \approx 3.998\,, \tag{6}$$

$$\mathcal{M} = \frac{\beta_c}{2} = \frac{3\varphi+\pi}{4} \approx 1.9989\,, \tag{7}$$

$$M \approx M_{\rm Pl} \quad \text{(free EFT parameter; we adopt } M = M_{\rm Pl} \text{ as reference)}\,. \tag{8}$$

The parameter β_c encodes the strength of the *disformal* coupling of this paper, and it is now the only β_c in the series. **Note on an earlier cross-reference:** previous versions related it to a conformal-sector coupling $\beta_c = -\mathcal{A}$ in Paper 7, presenting the two as sign conventions of one constant. That conformal coupling is withdrawn (Paper 7): the shooting that extracted it normalised the field density to the saturation $0.839950 = |w_0|$ rather than the density fraction $1-\Omega_m = 0.691119$, and with the correct target it changes sign. There is therefore no longer a second β_c to reconcile, and the disformal coupling

used here stands on its own. We adopt $\beta_c \equiv +\mathcal{A} > 0$, of which only the magnitude $|\beta_c| = \mathcal{A} = 3.997847$ enters the deflection angle.

For the purposes of this paper, we work in the quasi-static, non-relativistic limit ($v \ll c$, $\Phi_N \ll 1$), which is appropriate for galaxy and cluster scales where $k \gg aH$.

3 Scalar Field Equation and Gradient Closure

3.1 Quasi-static limit

Varying the action (2) with respect to ϕ gives the scalar equation of motion. In the quasi-static approximation ($|\dot{\phi}| \ll |\nabla\phi|$, $|\ddot{\phi}| \ll |\nabla^2\phi|$) and at leading order in the kinetic expansion, the non-linear kinetic term X^2/M^4 is subdominant on sub-Hubble scales. The equation reduces to

$$\frac{1}{\mathrm{KAL}_0}\,\nabla^2\phi \;=\; \frac{\beta_c}{M_{\rm Pl}}\,\rho_{\rm DM}\,, \tag{9}$$

where the right-hand side arises from the disformal coupling of dark matter to $\tilde{g}_{\mu\nu}$: dark matter acts as an effective source for ϕ with coupling strength $\beta_c/M_{\rm Pl}$.

The Newtonian gravitational potential Φ_N satisfies the standard Poisson equation

$$\nabla^2\Phi_N \;=\; \frac{\rho_{\rm DM}+\rho_b}{2M_{\rm Pl}^2}\,, \tag{10}$$

where we use units $8\pi G = 1/M_{\rm Pl}^2$.

3.2 Gradient closure

In the dark-matter-dominated regime ($\rho_{\rm DM} \gg \rho_b$), the Poisson equation (10) gives $\rho_{\rm DM} = 2M_{\rm Pl}^2\,\nabla^2\Phi_N$. Substituting this into the scalar equation (9) yields a relation between the Laplacians,

$$\nabla^2\phi \;=\; 2\mathrm{KAL}_0\,\beta_c\,M_{\rm Pl}\,\nabla^2\Phi_N\,, \tag{11}$$

that is, $\nabla^2(\phi - 2\mathrm{KAL}_0\,\beta_c\,M_{\rm Pl}\,\Phi_N) = 0$. The bracketed quantity is therefore harmonic. Imposing the boundary conditions appropriate to an isolated lens — both ϕ and Φ_N vanish at spatial infinity and the configuration contains no source-free harmonic component — the harmonic residual reduces to a constant, whose gradient vanishes. This yields the gradient closure relation

$$\boxed{\nabla\phi \;=\; 2\mathrm{KAL}_0\,\beta_c\,M_{\rm Pl}\,\nabla\Phi_N\,.} \tag{12}$$

This is the central structural relation of Paper 8: the scalar gradient tracks the Newtonian potential gradient with an algebraic coefficient $2\mathrm{KAL}_0\,\beta_c \approx 2 \times 5.521 \times 3.998 \approx 44.15$.

Equation (12) implies that wherever there is a gravitational potential well, the scalar field ϕ builds up a corresponding profile. Given the stated boundary conditions, this follows from the disformal coupling structure in the quasi-static limit rather than being imposed independently. We note that this derivation assumes $\rho_{\rm DM} \gg \rho_b$; in systems with significant baryonic fraction, equation (9) should be replaced by the full source including baryons, with a smaller effective coupling.

3.3 Scalar force on dark matter

The effective force on a dark matter particle in this scalar background is

$$F_\phi \;=\; -\frac{\beta_c}{M_{\rm Pl}}\,\nabla\phi \;=\; -2\mathrm{KAL}_0\,\beta_c^2\,\nabla\Phi_N\,. \tag{13}$$

The total force on a dark matter test particle is $F_{\rm tot} = F_N + F_\phi = -(1+2\mathrm{KAL}_0\beta_c^2)\,\nabla\Phi_N$, corresponding to an effective gravitational enhancement of $G_{\rm eff}/G = 1 + 2\mathrm{KAL}_0\beta_c^2 \approx 1 + 2(5.521)(3.998)^2 \approx 177$. This linear estimate belongs to the alternative MOND-like limit (Section 4); in the canonical limit the EFT structure $\alpha_B = \alpha_M = 0$ sets $\mu - 1 = 0$ (Section 6.3) and no such enhancement is observable.

4 Disformal Null Geodesic and Photon Deflection (Alternative MOND-like limit)

Scope of this section

The derivation in §4–5 assumes the *alternative MOND-like limit* in which the dark-matter distribution is sourced entirely by a scalar-force enhancement of the baryonic seed ($M_{\rm DM,eff} \propto \beta_c \cdot M_{\rm bary}$, no independent DM component). This is *not* the canonical SSEE model. The canonical canonical limit (single matter component $\Omega_m = 0.308881$, plus EFT $\alpha_B = \alpha_M = \alpha_T = 0$) is treated in §6–§6.3 and yields $\theta_{\rm E}^{\rm SSEE} \approx \theta_{\rm E}^{\rm GR-with-DM}$ (no $\sqrt{\mathcal{A}}$ or $\mathcal{M}$ amplification). The two limits make incompatible physical assumptions; the present section is retained as a derivation of pedagogical and historical interest only. Falsifiability statements derived from this section (§8) are accordingly conditional on the alternative limit being realised, which the canonical model does not predict.

4.1 Setup: photon propagation

Baryons and photons couple to the physical metric $g_{\mu\nu}$, not the disformal metric $\tilde{g}_{\mu\nu}$. A photon therefore follows the null geodesic of $g_{\mu\nu}$ in the weak-field perturbed Friedmann background:

$$ds^2 \;=\; -(1+2\Phi_N)\,dt^2 + (1-2\Phi_N)\,\delta_{ij}\,dx^i\,dx^j\,. \tag{14}$$

Standard GR light deflection by a point mass M gives the Einstein angle

$$\hat{\alpha}_{\rm GR} \;=\; \frac{4GM}{c^2 b}\,, \tag{15}$$

where b is the impact parameter.

4.2 Modified gravitational potential

Dark matter in SSEE does not follow purely Newtonian gravity; it distributes in response to the combined force $F_{\rm tot} = F_N + F_\phi$. For an isolated dark-matter-dominated system (galaxy cluster, compact group), the self-consistent equilibrium configuration produces an *effective* Newtonian potential $\Phi_{\rm eff}$ that satisfies

$$\nabla^2\Phi_{\rm eff} \;=\; \frac{\rho_{\rm DM,eff}}{2M_{\rm Pl}^2}\,, \tag{16}$$

where $\rho_{\rm DM,eff}$ is the dark matter density redistributed under the scalar force.

The effective lensing mass fraction is the full algebraic matter density, $\Omega_{m,\rm eff} = \Omega_m = \omega_m/h^2 = 0.308881$, at every scale — within 2% of the ΛCDM value 0.315. *(Earlier versions split this between a CDM and a ϕ-DM sector on either side of a free-streaming scale; that split is retracted with the particle, Paper 6, and the density is single and unpartitioned.)* (An earlier version justified this split by the "Two-Ω_m Criterion" of Paper 1. That criterion is itself withdrawn — $1 + w_0$ was never a matter density — so there is no split: the full $\Omega_m = 0.308881$ lenses at every scale.) The Einstein radius matches the GR-with-dark-matter prediction on all scales (see Section 7).

4.3 Total deflection angle in the linear regime

In the linear kinetic regime the scalar force operates at full strength. The total deflection of a photon passing an isolated dark-matter halo of effective gravitational mass $M_{\rm eff}$ at impact parameter b is, following the standard geodesic perturbation calculation (Will, 1993; Bartelmann & Schneider, 2001):

$$\hat{\alpha}_{\rm SSEE} = \frac{4GM_{\rm eff}}{c^2 b} \approx \beta_c \frac{4GM_{\rm bary}}{c^2 b}, \tag{17}$$

where the second equality holds in the limit where the dark-matter distribution is entirely sourced by the scalar-force-enhanced coupling to the baryonic seed $M_{\rm bary}$, and $\beta_c \approx 3.998$ is the disformal coupling.

Caveat: Equation (17) represents a working phenomenological estimate. A full solution requires self-consistent integration of the dark-matter distribution under both F_N and F_ϕ, which we defer to future N-body simulations. The result (17) should be understood as an upper bound on the lensing enhancement, applicable to systems dominated by scalar-force-driven dark matter accumulation.

Figure withdrawn

A figure here plotted the disformal lensing enhancement $\theta_E^{\rm SSEE}/\theta_E^{\rm GR}$ as a function of wavenumber, with a transition at a ϕ-DM free-streaming scale: below it the ratio approached $\sqrt{\beta_c} = \sqrt{\mathcal{A}} \approx 1.9995$, above it lensing recovered the GR baseline. **The figure is withdrawn**, because the scale that produced its entire k-dependence is retracted with the particle (Paper 6) and the matter sector is single and unpartitioned — there is no such transition to plot. The pedagogical limit (a) and the canonical limit (b) remain as analytic statements (§6.3); what is gone is the scale separating them.

5 Lensing Factor $\sqrt{\mathcal{A}}$ in the Alternative Limit (and Its Numerical Near-Coincidence with $\mathcal{M}$)

Naming clarification

The quantity derived in this section as the lensing factor is $\sqrt{\beta_c} = \sqrt{\mathcal{A}} \approx 1.99946$ (an exact algebraic expression). This is *not* the same as $\mathcal{M} = \mathcal{A}/2 \approx 1.9989$, which uses a different algebraic operation (division by 2 rather than square root). The two quantities agree to 0.03% because $\mathcal{A} \approx 4$, which makes both $\sqrt{\mathcal{A}}$ and $\mathcal{A}/2$ land near 2. We retain the comparison below for completeness, but the *lensing factor in the alternative limit is* $\sqrt{\mathcal{A}}$*, not* $\mathcal{M}$. Earlier drafts of this paper conflated the two under the label "$\mathcal{M}$ emerges in lensing"; the present version corrects that overstatement. In the canonical limit (§6–§6.3) neither $\sqrt{\mathcal{A}}$ nor $\mathcal{M}$ amplifies the observable Einstein radius — the prediction is GR lensing with realistic DM halo.

5.1 Einstein radius in SSEE

The Einstein ring is the locus where a source is perfectly aligned with the lens and observer. The Einstein radius $\theta_{\rm E}$ satisfies the lens equation

$$\theta_{\rm E} \;=\; \hat{\alpha}(\theta_{\rm E})\,\frac{D_{\rm LS}}{D_S}\,, \tag{18}$$

where D_L, D_S, $D_{\rm LS}$ are angular diameter distances. Substituting (17) with $b = D_L\,\theta_{\rm E}$:

$$\theta_{\rm E}^2 \;=\; \beta_c\,\frac{4GM_{\rm bary}}{c^2}\,\frac{D_{\rm LS}}{D_L D_S}\,. \tag{19}$$

Comparing to the GR result $(\theta_{\rm E}^{\rm GR})^2 = 4GM_{\rm bary}\,D_{\rm LS}/(c^2 D_L D_S)$, we obtain

$$\boxed{\frac{\theta_{\rm E}^{\rm SSEE}}{\theta_{\rm E}^{\rm GR}} \;=\; \sqrt{\beta_c} \;=\; \sqrt{\mathcal{A}} \;\approx\; 1.9995\,.} \tag{20}$$

5.2 Algebraic identification with $\mathcal{M}$

The structural constant $\mathcal{M}$ is defined in equation (1):

$$\mathcal{M} \;=\; \frac{3\varphi + \pi}{4} \;=\; \frac{\beta_c}{2} \approx 1.9989\,. \tag{21}$$

Meanwhile, $\sqrt{\beta_c} = \sqrt{2\mathcal{M}}$. At the specific value $\mathcal{M} \approx 2$, we have $\sqrt{2\mathcal{M}} \approx \sqrt{4} = 2 \approx \mathcal{M}$, and more precisely:

$$\sqrt{\beta_c} \;=\; \sqrt{2\mathcal{M}} \;\approx\; 1.9995\,, \qquad \mathcal{M} \;\approx\; 1.9989\,, \tag{22}$$

so $\sqrt{\beta_c}/\mathcal{M} = 1.00027$ (agreement to 0.03%).

This near-coincidence $\mathcal{M} \approx \sqrt{\beta_c}$ is a consequence of the structural postulate $\beta_c \equiv \mathcal{A} = 2\mathcal{M}$, combined with $\mathcal{M} \approx 2$. Concretely:

$$\mathcal{M} = 2 \iff \beta_c = 4 \iff \sqrt{\beta_c} = 2 = \mathcal{M}\,. \tag{23}$$

The algebraic values give $\mathcal{M} = 1.9989$ and $\beta_c = 3.997847$, both differing from the integer approximation by $< 0.06\%$.

Therefore, to within 0.03%, the lensing amplification factor coincides with the CMB horizon mapping factor:

$$\theta_{\rm E}^{\rm SSEE} \;\approx\; \mathcal{M} \times \theta_{\rm E}^{\rm GR}\,, \tag{24}$$

where the same $\mathcal{M}$ (the "Mirror Ratio") appears in:

- Paper 9: the Hubble screening fraction $f_{\rm scr}^{\rm IR} = s_K^{\rm IR}/(3\,\mathcal{M}) = (\pi - \varphi)/\Omega^2$
- This paper: $\theta_{\rm E}^{\rm SSEE} \approx \mathcal{M} \times \theta_{\rm E}^{\rm GR}$

(Under the ω_m-direct reframe, OP-8 dissolved, $\mathcal{M}$ no longer enters the CMB matter sector: $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.308881$ with no matter factor, and $r_d = 147.2\,$Mpc follows directly from ω_m, both superseding the earlier MIRA-mapped forms.)

This recurrence of $\mathcal{M}$ across the Hubble-screening and strong-lensing sectors reflects a single structural input: $\mathcal{M}$ is fixed algebraically by φ and π, not tuned. The lensing entry holds at the 0.03% level through the near-coincidence $\sqrt{\beta_c} \approx \mathcal{M}$ (Section 5), not as an exact identity.

6 Solar-System Compliance

6.1 Non-linear kinetic term

The X^2/M^4 term in the SSEE action (3) places SSEE in the *k-mouflage* class of kinetic screening theories (Brax & Valageas, 2014), distinct from Galileon (DGP) (Deffayet et al., 2001) or Vainshtein screening (Vainshtein, 1972; Babichev & Deffayet, 2013); for a review of scalar screening mechanisms see Koyama (2018). We do *not* quote a k-mouflage screening radius. Earlier versions of this paper did, from $r_{\rm km}^3 = M_{\rm obj}/(4\pi M_{\rm Pl} M^2)$; in natural units the right-hand side has dimension $[\text{mass}]^{-2} = [\text{length}]^2$, not $[\text{length}]^3$, so the expression is not dimensionally consistent and its numerical values changed by a factor 10^3 depending on whether masses were written in GeV or in eV. Those values are withdrawn. No screening radius is needed for the results of this paper: solar-system compliance and the canonical lensing prediction rest on two facts that hold for any value of M — baryons do not couple to the scalar (Section 6.2), and the EFT functions $\alpha_B = \alpha_M = \alpha_T = 0$ of Paper 7 set $\mu - 1 = 0$ (Section 6.3). The non-linear profile of the scalar around individual halos is left to a numerical calculation.

6.2 GR compliance from selective coupling

Solar-system gravitational tests are automatically satisfied by the *selective coupling* structure of the action (2):

- **Baryons** couple to the physical metric $g_{\mu\nu}$ only. They feel no direct force from the scalar gradient $\nabla\phi$.
- **Dark matter** couples to the disformal metric $\tilde{g}_{\mu\nu}$. It is the sole source of ϕ through the scalar EOM (9).
- In the solar system, the dark-matter density is $\rho_{\rm DM} \approx 0.3\,\text{GeV/cm}^3$, far below galactic overdensities. The scalar field is barely sourced locally, so $\nabla\phi \approx 0$ and $\tilde{g}_{\mu\nu} \approx g_{\mu\nu}$.

Therefore all standard solar-system tests (Cassini time delay, perihelion precession, lunar laser ranging (Will, 2014)) pass at leading order without requiring any non-linear screening. This mechanism operates independently of the value of M and holds in both the IR ($M = M_{\rm Pl}$) and UV ($M = 9.68\,\text{meV}$) limits.

6.3 EFT suppression of fifth-force corrections

Beyond selective coupling (Section 6.2), the Bellini-Sawicki structure of the SSEE EFT (Bellini & Sawicki, 2014; Almeida Vallejo, 2026c) provides a second independent suppression of any fifth-force correction at solar-system scales.

Paper 7 established that SSEE has (Almeida Vallejo, 2026c)

$$\alpha_B \;=\; \alpha_M \;=\; \alpha_T \;=\; 0\,, \tag{25}$$

with $\alpha_K = 15.591336$ as the sole non-vanishing EFT parameter. The Bellini-Sawicki effective Newton constant is (Bellini & Sawicki, 2014)

$$\mu - 1 \;=\; \frac{(\alpha_B + \alpha_M)^2}{\alpha_K} \;=\; 0\,, \tag{26}$$

and the gravitational slip parameter

$$\gamma_{\rm PPN} - 1 \;=\; -\frac{2\alpha_T}{1+\alpha_T} \;=\; 0\,. \tag{27}$$

Both vanish identically. In the quasi-static limit at sub-Hubble wavenumber k, the residual fifth-force ratio is further suppressed by $(H/k)^2$ (Bellini & Sawicki, 2014):

$$\left.\frac{F_\phi}{F_N}\right|_{\rm QS} \;\sim\; \frac{H^2}{k^2}\,|1 + w_{\rm DE}|\,\Omega_{\rm DE}\,. \tag{28}$$

At the Earth–Sun separation ($k \sim 1/\text{AU}$, $ck/H_0 \approx 9.2 \times 10^{14}$):

$$\left.\frac{F_\phi}{F_N}\right|_{1\,\text{AU}} \;\approx\; 1.6 \times 10^{-31} \;\ll\; 10^{-5} \quad \text{(Cassini bound (Will, 2014))}\,, \tag{29}$$

satisfying the solar-system constraint by a factor of 6×10^{25}.

The two GR-compliance mechanisms in SSEE are:

1. **Selective coupling (Section 6.2):** Baryons couple only to the physical metric $g_{\mu\nu}$, with no direct scalar force regardless of M or scale.
2. **EFT Bellini-Sawicki structure ($\alpha_B = \alpha_M = \alpha_T = 0$):** The effective Newton constant and gravitational slip both vanish exactly, suppressing any residual fifth force to $F_\phi/F_N \sim 10^{-31}$ at solar-system scales.

No non-linear screening radius is invoked: the EFT suppression in equation (29) is sufficient for all solar-system tests. The cosmological background Hubble flow remains in the linear kinetic regime ($X_{\rm bg}^2/M^4 \approx 5.7 \times 10^{-4}$, Paper 10 (Almeida Vallejo, 2026d)), so the background equations of motion are unaffected by the UV completion.

7 Strong Lensing Predictions

Gravitational lensing by dark matter halos was established observationally by Bradac et al. (2006) and Clowe et al. (2006) in cluster collisions; for the lensing formalism see Narayan & Bartelmann (1996) and Schneider, Ehlers & Falco (1992).

7.1 Scale-dependent lensing signature — *withdrawn*

Two subsections here derived a scale-dependent lensing signature from a ϕ-DM free-streaming scale: a lensing mass fraction $f_{\rm lens}(k)$ that stepped from 1 below $k_{\rm fs}$ to $\Omega_{\rm CDM}/\Omega_m$ above it, and the observational consequences that followed (an anomalous dynamical-to-lensing mass ratio, a transition in the radial Einstein-ring profile near $\ell_{\rm fs}$, and a suppression in galaxy–galaxy shear on 1–10 kpc scales).

All of it is withdrawn. The construction rested on the two-sector partition of Paper 6, which is retracted: the subtraction defining the second sector's density took a measured density minus $1 + w_0$, a number from the equation of state, and so defined nothing. With a *single* unpartitioned matter component, $\Omega_m = \omega_m/h^2 = 0.308881$, there is no free-streaming scale, no step in $f_{\rm lens}(k)$, and therefore no scale-dependent lensing signature to predict. $f_{\rm lens}(k) = 1$ at every k.

What survives is the canonical prediction itself, which never depended on the partition: with real clustering cold dark matter (ω_c, §1) and $\alpha_B = \alpha_M = \alpha_T = 0$ from Paper 7, the observable lensing reduces to GR-with-a-dark-matter-halo at all scales.

7.2 Reference numbers

Table 1 summarises the key numerical predictions.

Table 1: SSEE strong-lensing predictions compared to GR/ΛCDM.

Observable	**GR/ΛCDM**	**SSEE**	**Scale**
$\theta_{\rm E}^{\rm SSEE}/\theta_{\rm E}^{\rm GR}$	1	1 (single matter sector, all k)	All
Two rows here previously split this ratio at a ϕ-DM free-streaming scale (1 below, $1/\sqrt{\mathcal{M}} \approx 0.707$ above); both are withdrawn with the partition — see §7.1.			
$M_{\rm dyn}/M_{\rm lens}$ ratio	1 (GR)	> 1 (fifth force on DM)	Linear regime
GW speed anomaly (α_T)	0	0 (exact)	All

7.3 Worked numerical example: cluster A1689

As a concrete numerical test discriminating between the two limits of Section 8, we compute the deflection angle for cluster A1689, one of the most powerful gravitational lenses known, using the following reference values: total projected mass within the Einstein radius $M_{\rm tot} \approx 2.7 \times 10^{14}\, M_\odot$ (Limousin et al., 2007), baryonic mass (gas + stars, $f_b \approx 0.14$) $M_{\rm bary} \approx 3.8 \times 10^{13}\, M_\odot$, effective lensing impact parameter $b = 250$ kpc (corresponding to the observed Einstein radius $\theta_{\rm E} \approx 45''$ at $z_L = 0.183$, $z_S = 1.0$).

Using $G = 4.302 \times 10^{-3}\,\mathrm{pc}\,M_\odot^{-1}\,(\mathrm{km/s})^2$ and $c = 2.998 \times 10^5\,\mathrm{km/s}$:

$$\hat{\alpha}_{\mathrm{canonical}} = \frac{4GM_{\mathrm{tot}}}{c^2\,b} = \frac{4 \times 4.302 \times 10^{-3} \times 2.7 \times 10^{14}}{(2.998 \times 10^5)^2 \times 250 \times 10^3} \approx 2.07 \times 10^{-4}\,\mathrm{rad} = 42.6'', \quad (30)$$

$$\hat{\alpha}_{\mathrm{GR}}^{\mathrm{bary}} = \frac{4GM_{\mathrm{bary}}}{c^2\,b} \approx 2.9 \times 10^{-5}\,\mathrm{rad} = 6.0'', \quad (31)$$

$$\hat{\alpha}_{\mathrm{alt}} = \beta_c\,\hat{\alpha}_{\mathrm{GR}}^{\mathrm{bary}} = 3.997847 \times 2.9 \times 10^{-5} \approx 1.16 \times 10^{-4}\,\mathrm{rad} = 24.0''. \quad (32)$$

The canonical limit (EFT B–S suppression, lensing sourced by the full matter density $\Omega_m = 0.308881$, §6.3) reproduces the observation: $\hat{\alpha}_{\mathrm{canonical}} = 42.6''$ versus the observed $\theta_{\mathrm{E}} \approx 45''$, agreement at the 5% level (the small residual is absorbed by the projection geometry and the mass-profile modelling of Limousin et al., 2007). The alternative MOND-like limit, in which lensing is sourced by baryons enhanced by $\beta_c \approx 4$, falls short of the observation by a factor ≈ 1.9 — precisely the well-known residual missing-mass problem that MOND-type theories face in galaxy clusters. A1689 therefore acts as a discriminating test between the two limits and selects the canonical one, consistent with the cold dark-matter density that SSEE predicts forward, $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ (Paper 1).This example also demonstrates that Eq. (17) produces physically reasonable deflection angles, not merely formal algebraic expressions.

8 Falsifiable Predictions

Scope of falsifiability — revised 2026-08

The predictions enumerated below are pre-committed falsifiable observables of the canonical model. An earlier version justified the selection of the canonical Bellini–Sawicki limit (b) by appealing to a ϕ-dark-matter particle mass fixed in Paper 6. **Paper 6 has retracted that particle**, so that justification is withdrawn — but the selection stands on two legs neither of which the particle ever supplied: the model carries real clustering cold dark matter ($\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, OP-8), and the fifth force is suppressed identically by $\alpha_B = \alpha_M = 0$ (Paper 7), giving $\mu - 1 = (\alpha_B + \alpha_M)^2/\alpha_K = 0$. The prediction is therefore retained as unconditional, on firmer ground than before. The two-limit analysis is kept in full because limit (a) is a useful falsifiable contrast, not because the choice is open.
Two entries in the original list — the $P(k)$ free-streaming cutoff and the scalar mass itself — are *withdrawn* rather than revised, and are marked as such where they appeared. They were properly falsifiable; that was never the difficulty. The difficulty is that falsifiability is the minimum requirement for a hypothesis to be scientific, **not** evidence that the entity it concerns exists. Publishing a bound on a particle whose defining construction has been withdrawn would grant it existence through the back door, so the entries are removed rather than weakened.

We collect the pre-committed falsifiable predictions of this paper. A single decisive measurement inconsistent with any of these would falsify the corresponding sector of the SSEE framework.

1. **Free-streaming cutoff — *withdrawn.*** A suppression of $P(k)$ above a ϕ-DM free-streaming scale was previously listed here as a falsifier. It is withdrawn together

with the particle (Paper 6): the second sector was defined by a subtraction that mixed a measured density with an equation-of-state number, so there is no free-streaming species to look for. The growth-sector falsifier is now the shear amplitude itself, $S_8 = 0.7559 \pm 0.0189$[1] against raw KiDS-1000 $\xi_\pm$ data.

2. **Lensing mass transition (canonical limit prediction)**: In the canonical single-sector model with EFT B-S structure (§6.3), the predicted observable lensing is $\theta_{\rm E}^{\rm SSEE} \approx \theta_{\rm E}^{\rm GR-with-DM-halo}$ at *all* scales accessible to JWST/HST strong lensing, because the residual fifth-force is suppressed to $F_\phi/F_N \sim 10^{-15}$ at sub-Mpc scales. This is consistent with current SLACS, BELLS, and BELLS GALLERY samples (mean total density slope $\gamma = 2.078 \pm 0.027$, Auger et al. 2010; $\mathrm{M_E/M_{dyn}} \approx 1.21$ for SIS modelling, Li et al. 2018 — both consistent with standard NFW+stellar models within errors). *Falsification criterion*: If JWST finds a systematic $\theta_{\rm E}$ enhancement of order $\sqrt{\mathcal{A}} \approx 2$ relative to ΛCDM in any clean sub-Mpc sample, the canonical limit would be falsified *and* the alternative MOND-like limit of §4–5 would be the physically realised scenario. Current data does not show such an enhancement; the canonical prediction stands. [2]

3. **Dynamical vs. lensing mass discrepancy (canonical limit)**: In the canonical limit, the EFT suppression of §6.3 predicts $M_{\rm dyn}/M_{\rm lens} \approx 1$ at galaxy and cluster scales, consistent with the SLACS/BELLS observation $M_E/M_{\rm dyn} \approx 1.21$ for SIS modelling (Li et al., 2018) and within 1σ of unity for power-law profiles. The factor $\beta_c \approx 4$ enhancement applies only in the alternative MOND-like limit (§4–5) and is not the canonical prediction. *Falsification criterion*: If a clean sample shows $M_{\rm dyn}/M_{\rm lens}$ systematically $\gtrsim 2$ across $\gtrsim 100$ clusters, the canonical limit is falsified and the alternative limit would be selected.

4. **Gravitational wave speed**: $\alpha_T = 0$ (exact), same as GR. The GW170817 constraint $|\alpha_T| < 10^{-15}$ (LIGO/Virgo/Fermi, 2017) is already satisfied. *Falsification criterion*: Any detection of $\alpha_T \neq 0$ with current or future GW detectors would require modification of the SSEE gravitational sector.

5. **Scalar mass m_ϕ — *withdrawn* (Paper 6).** This entry previously gave an algebraic ϕ-DM mass and its free-streaming falsifier. Both are retracted. Falsifiability was never the problem with that entry; the problem is that the object it constrained does not exist, and a bound on a non-existent particle would grant it existence through the back door.

[1] Updated 2026-09-20. Against the later KiDS-Legacy release (357 raw $\xi_\pm$ points; Wright et al. 2025, arXiv:2503.19441 — published sixteen months before this analysis, over which no temporal priority is claimed) the amplitude the shear asks for lies 0.49σ from the one the CMB fixes under the *same* algebraic background, so A_s need not be free either: fixed to that value, the growth sector carries *no* free cosmological parameter and predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 measured. The three χ^2 compared span 1.0 over 357 points, so the shear does *not* discriminate between the models; what separates them is the parameter count. See Paper 6.

[2] Earlier drafts of this paper presented a $\sim 30\%$ deficit prediction at $k > k_{\rm fs}$, derived from a two-sector free-streaming argument applied to the alternative-limit lensing factor. That deficit prediction is withdrawn: the two-sector suppression argument applies to the matter power spectrum (where the prediction is given by Item 1 above) but does not transfer directly to the observable Einstein radius once the EFT suppression of §6.3 is correctly applied. See OPEN_PROBLEMS.md OP-13.

9 Discussion

9.1 Where $\mathcal{M}$ does and does not appear

$\mathcal{M} = (3\varphi + \pi)/4 = 1.998924$ appears in exactly one place in the current framework: the Hubble-screening amplitude of Paper 9,

$$f_{\rm screen}^{\rm IR} = \frac{s_K^{\rm IR}}{3\,\mathcal{M}}, \qquad s_K^{\rm IR} = -3w_0(1+w_0) = 0.403302, \tag{33}$$

where s_K is a pressure-dilution rate carrying only the equation of state. Earlier versions of this subsection listed two further appearances, and both are withdrawn:

- $r_{d,\rm eff} = r_{d,\rm SSEE}/\mathcal{M}$. This mapping existed to reconcile $r_d = 175.6$ Mpc with the Planck-inferred 147.09 Mpc. That 175.6 was an artefact of putting $\Omega_{m,\rm dyn} = 0.160050$ — which is $1+w_0$, a quantity of the equation of state — into the background geometry, where the total matter density belongs. With $\Omega_m = \omega_m/h^2 = 0.308881$ the model gives $r_d = 147.17$ Mpc directly (CAMB), 1.0005× the ΛCDM value at the Planck 2018 parameters, and there is nothing left for $\mathcal{M}$ to correct.

- $\Omega_{m,\rm CMB} = \mathcal{M}\,\Omega_{m,\rm dyn}$. There is no matter factor. The CMB matter density is the forward identity $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ of Paper 1, and $\Omega_m = \omega_m/h^2$ is derived from it; $\Omega_{m,\rm dyn}$ is not a density and cannot stand on the other side of such a relation. OP-8 closed on this point.

In the alternative MOND-like lensing limit (§4–5) the cantidad that emerges is $\sqrt{\mathcal{A}} \approx 1.99946$, which is *numerically close* to $\mathcal{M}$ to within 0.03% but is a different algebraic operation on $\mathcal{A}$ ($\sqrt{\mathcal{A}}$ vs $\mathcal{A}/2$). The near-coincidence is a consequence of $\mathcal{A} \approx 4$, which makes both $\sqrt{\mathcal{A}}$ and $\mathcal{A}/2$ land near 2; it is not a structural identity. We therefore do *not* count the lensing entry as a third independent appearance of $\mathcal{M}$.

In the canonical limit (§6–§6.3) neither $\sqrt{\mathcal{A}}$ nor $\mathcal{M}$ enters the observable Einstein radius: the EFT suppression gives $\theta_{\rm E}^{\rm SSEE} \approx \theta_{\rm E}^{\rm GR-with-DM}$.

Summary: $\mathcal{M}$ retains its algebraic value $(3\varphi+\pi)/4$ and enters the framework through $f_{\rm screen}$ alone. Its appearance in the lensing sector of earlier drafts was a name-conflation of $\sqrt{\mathcal{A}}$ with $\mathcal{M}$ via numerical near-coincidence; its appearances in the CMB and LSS sectors rested on a geometry that used $1 + w_0$ where a density belongs. All three are withdrawn. What $\mathcal{M}$ *is*, physically, is an open question and not a filled one.

9.2 Limitations and open questions

The derivations in this paper adopt the quasi-static, non-relativistic limit and treat the scalar force in the linear kinetic regime. Several steps remain to complete the strong-gravity picture:

1. **Self-consistent DM distribution**: The estimate $\hat{\alpha}_{\rm SSEE} \approx \beta_c\,\hat{\alpha}_{\rm GR}$ assumes the dark-matter distribution is entirely determined by the scalar force. A proper calculation requires solving the coupled DM–scalar system numerically (N-body simulation with scalar force module).

2. **Non-linear field profile**: No k-mouflage screening radius is quoted (Section 6); the non-linear profile of the scalar around individual halos requires a numerical calculation. None of the predictions of this paper depends on it.

3. **Relativistic corrections**: On cluster scales at $z \sim 0.5$, relativistic corrections to the geodesic ($\Phi_N \sim 10^{-4}$–10^{-3}) are small but not negligible at the precision of JWST.

4. **Baryonic feedback**: The lensing predictions in Table 1 are at the linear DM level. Baryonic feedback redistributes mass on sub-100 kpc scales and will modify the predicted Einstein ring profiles.

9.3 Connection to UV completion

The X^2/M^4 kinetic term in the SSEE action is the leading non-linear operator in a derivative expansion. Paper 10 of the SSEE series derives $M^4 = 45\alpha^2\,\rho_{\rm c,0} = 5\varphi^8\,\rho_{\rm c,0}$ from the inflationary α-attractor parameter $\alpha = \varphi^4/3$, fixing

$$M \;=\; \varphi^2\,5^{1/4}\,\rho_{\rm c,0}^{1/4} \;\approx\; 9.68\ \text{meV} \tag{34}$$

algebraically without free parameters. This is a *cosmological-scale* cutoff—thirty orders of magnitude below $M_{\rm Pl}$—so no quantum-gravity UV completion is required. The EFT derivative expansion is valid over the entire cosmological regime; the X^2/M^4 operator is a dark-energy self-interaction, not a near-Planck correction.

10 Conclusions

We have derived the strong-gravity consequences of the SSEE disformal coupling and found:

1. **Gradient closure**: In the quasi-static limit, the scalar field profile tracks the Newtonian potential via $\nabla\phi = 2\mathrm{KAL}_0\beta_c M_{\rm Pl}\,\nabla\Phi_N$, with $\mathrm{KAL}_0 \approx 5.521$ and $\beta_c \approx 3.998$ fixed algebraically.

2. **Lensing amplification**: The total deflection angle is amplified by β_c relative to the GR baryonic-mass prediction, yielding $\theta_{\rm E}^{\rm SSEE} \approx \sqrt{\beta_c} \approx \mathcal{M}$ times the GR Einstein radius.

3. **MIRA emergence**: The structural constant $\mathcal{M} = (3\varphi + \pi)/4 \approx 1.9989$, which enters the Hubble screening amplitude of Paper 9, also matches the lensing amplification of the alternative limit to within 0.03% (via the near-coincidence $\sqrt{\beta_c} \approx \mathcal{M}$). The former "two-sector mass ratio" reading is withdrawn with the partition.

4. **GR compliance**: Solar-system tests are satisfied by two independent mechanisms: selective coupling, under which baryons couple only to the physical metric $g_{\mu\nu}$ (Section 6.2), and the Bellini-Sawicki EFT structure $\alpha_T = \alpha_M = \alpha_B = 0$ from Paper 7, which suppresses any residual fifth force to $F_\phi/F_N \sim 10^{-31}$ at 1 AU (Section 6.3).

5. *(A two-sector scale-dependent lensing signature was listed here; it is retracted with the ϕ-DM particle, Paper 6.)*

Paper 8 thus extends the SSEE predictive reach from the linear cosmological regime into the non-linear strong-gravity regime, while maintaining the zero-free-parameter structure of the algebraic framework.

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